

Detecting Convolutional Codes via a Markovian Statistic

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Abstract—Identifying the unknown channel code corresponding to the given intercepted data is an important problem in military surveillance and in wireless communication. In this paper, we focus on the situation when the given data corresponds to either of the two potential convolutional codes and the goal is to detect the correct code. While a variety of code detection algorithms are available in the literature, the key contribution of our work lies in the novel solution and the corresponding analysis. We decode the received data via Viterbi algorithm using either of the code and consider the corresponding sequence of Viterbi metric for code detection. We identify that the sequence of Viterbi metric follows a Markov chain (MC) and formulate the code detection problem as a hypothesis testing between two MCs. We provide a method to find the state-transition matrices for the two MCs and study the analytical asymptotic performance via the associated error exponent. We validate the performance of our method via a variety of simulations and also compare our performance with the method available in the literature.

Index Terms—Convolutional codes, Blind reconstruction of channel codes, Hypothesis testing between Markov chains

I. INTRODUCTION

In digital communication systems, channel coding is essential for reliable data transmission over noisy communication channels. Convolutional codes are a widely used class of channel codes. Channel codes encode input data streams into codewords, making communication systems more robust to noise and errors. However, in certain scenarios, it becomes essential to detect the encoding scheme used, particularly when the encoder is not explicitly known to the observer. This capability is critical in adaptive modulation and coding (AMC) frameworks [1], where systems dynamically adjust modulation and coding schemes to optimize spectral efficiency under changing channel conditions. Efficient encoding scheme detection enables receivers to accurately decode transmitted messages without requiring additional signaling, minimizing overhead [2]. Additional applications include military surveillance for signal intelligence and cognitive radio networks, where detecting primary users' encoding minimizes interference [3].

The identification of unknown convolutional codes has been extensively studied [2], [4]–[10], along with other families such as cyclic [11]–[14], LDPC [15]–[18], Turbo codes [7], [19], learning-based approaches [20] and hybrid architectures [21].

In this work, we focus on the situation in which the candidate convolutional codes are known to the receiver¹. The code identification problem when the set of candidate codes are known is studied in [2], [22]–[25]. While [2] considers the situation where the candidate codes are convolutional codes, [22]–[25] consider the situation when the candidate codes are linear block codes.

In this work, we focus on the situation where the transmitter uses convolutional code-1 or code-2 and the aim is to identify the correct code that corresponds to the given noise-affected intercepted data (see Fig. 1). Our contributions are summarized as follows:

- We provide a novel solution to this problem in which we decode the intercepted data using the Viterbi algorithm using either code and use the corresponding Viterbi metric as a *test statistic* for code detection.
- We identify that the sequence of Viterbi metric follows a Markov chain (MC) whose state transition matrix is different for different convolutional codes. This allows us to formulate and solve the code detection problem as a hypothesis testing between two MCs (Section III-A).
- We study the asymptotic performance of our method by estimating the error exponent associated with the hypothesis testing. We find this error exponent analytically and also validate it empirically (Table II).
- We establish validity of our method via a variety of simulations and also compare our performance with the method [2] available in the literature (Fig. 3).

Organization: The code detection problem studied in this work is summarized in Section II. The proposed approach and the corresponding simulation results are provided in Sections III and IV respectively. Finally, the conclusion and future work is discussed in Section V.

Notation: Let $\mathbb{F}_2$ denote the binary field with elements 0 and 1. A vector is denoted by a boldface letter and its components are denoted by small-case letters. For example, vector $\mathbf{v} = [v_0 \ v_1 \ \dots \ v_{n-1}]$. Random variables (and vectors) are denoted by upper-case letters and their realizations are denoted by the lower-case letters. For example, $\mathbf{v}$ is a realization of a random vector $\mathbf{V}$.

¹In wireless communication, such setup arises in AMC, where the set of channel codes of the transmitter is known to the receiver [1]–[3].

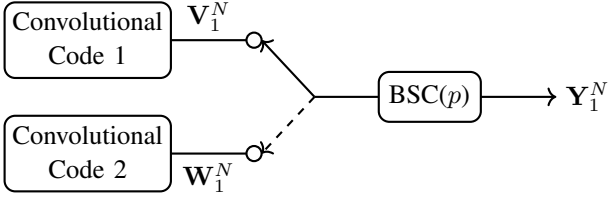


Fig. 1: System model for our method

II. SYSTEM MODEL

The system model for the problem of distinguishing between the two convolutional codes is illustrated in Fig. 1. We consider the situation when the transmitter uses either convolutional code-1 or code-2 and the goal is to identify the correct code corresponding to the given noise-affected received sequence. Let $\mathcal{C}(k, n, m)$ denotes the convolutional code of rate k/n and memory m . A convolutional code is described via the generator sequences denoted by the set of vectors $\mathbf{g}^{(0)}, \mathbf{g}^{(1)}, \dots, \mathbf{g}^{(n-1)}$ where each $\mathbf{g}^{(i)} \in \mathbb{F}_2^{(m+1)}$ [26, Ch. 11]. The generator sequences are represented using their respective octal representations. In this work, we assume that both the codes have the same parameters k, n , and m but have different generator sequences. We also assume that both the codes are feed-forward convolutional codes. For the sake of simplicity of notation, we ignore the parameters k, n , and m from the notation $\mathcal{C}(k, n, m)$ of a convolutional code and denote it by $\mathcal{C}$. The two potential codes used by the transmitter are then denoted by $\mathcal{C}_1$ and $\mathcal{C}_2$.

Let $\mathbf{U}_1^N = \mathbf{U}_1, \mathbf{U}_2, \dots, \mathbf{U}_N$ be the input message sequence, where $\mathbf{U}_t \in \mathbb{F}_2^k$ is the input at t -th time instant for $t = 1, 2, \dots, N$. We assume that each input message bit is chosen independently according to the Bernoulli(0.5) distribution. This input sequence is either encoded by code $\mathcal{C}_1$ or $\mathcal{C}_2$. While the encoded sequence of code $\mathcal{C}_1$ is denoted by $\mathbf{V}_1^N = \mathbf{V}_1, \mathbf{V}_2, \dots, \mathbf{V}_N$, the encoded sequence of code $\mathcal{C}_2$ is denoted by $\mathbf{W}_1^N$, where $\mathbf{V}_t, \mathbf{W}_t \in \mathbb{F}_2^n$. We assume that the transmitter uses both the codes with equal probability.

The encoded sequence of either $\mathcal{C}_1$ or $\mathcal{C}_2$ is transmitted via the binary symmetric channel (BSC) of crossover probability p to receive the sequence $\mathbf{Y}_1^N$. Let $\mathbf{E}_1^N$ denotes the random vector corresponding to the error introduced by the BSC(p). For the given received sequence $\mathbf{y}_1^N$, our goal is to determine the convolutional code corresponding to it. We formulate this code detection problem as a binary hypothesis testing problem where the two hypotheses H_1 and H_2 are given by

$$\begin{aligned} H_1 : \mathbf{Y}_1^N &= \mathbf{V}_1^N + \mathbf{E}_1^N \\ H_2 : \mathbf{Y}_1^N &= \mathbf{W}_1^N + \mathbf{E}_1^N. \end{aligned} \quad (1)$$

Note that since the transmitter uses both the codes with equal probability, hypotheses H_1 and H_2 are equally likely.

III. PROPOSED METHOD

In this section, we propose a new method for the problem of distinguishing between the two convolutional codes (see

Eq. (1)). This section is divided into three subsections. In Section III-A, we first discuss our method for code detection. The key idea of our method is to reduce the hypothesis problem of Eq. (1) to the problem of hypothesis testing between two Markov chains (MCs). The details related to obtaining the state transition matrices associated with the underlying MCs are discussed in Section III-B. Finally in Section III-C, we study the asymptotic performance of the hypothesis testing via the *error exponent*, which characterizes the rate with which the error in the hypothesis testing will go to zero as N tends to infinity [27], [28].

A. Proposed method for code detection

For the code detection problem of Eq. (1), one can use the optimal likelihood ratio tests (LRT), where one needs to compute the likelihood of observing $\mathbf{Y}_1^N$ under hypothesis H_j , denoted by $\mathbb{P}(\mathbf{Y}_1^N | H_j)$ for $j = 1, 2$. While the computation of $\mathbb{P}(\mathbf{Y}_1^N | H_j)$ can be done efficiently using the BCJR algorithm [22], [29], analyzing the theoretical performance guarantees of the associated LRT might not be trivial. Hence in the literature, researchers have come up with various *suboptimal statistics* of the received data $\mathbf{Y}_1^N$ that allows one to distinguish between the two codes, while enabling the analytical and numerical study of the underlying hypothesis testing setup.

In this work, we propose to use a new statistic of the received data $\mathbf{Y}_1^N$ that is obtained in an intermediate step of the Viterbi decoding algorithm [26], [30]. To get this statistic, without loss of generality, we always decode the received data $\mathbf{Y}_1^N$ using code $\mathcal{C}_1$ via the Viterbi algorithm. In an intermediate step of the Viterbi algorithm, one computes the sequence of *relative Viterbi metric*, denoted by $\mathbf{D}_0, \mathbf{D}_1, \dots, \mathbf{D}_N$, where each $\mathbf{D}_t$ is a vector of length 2^m such that each component in $\mathbf{D}_t$ corresponds to a vertex in the trellis at time t . At time $t = 0$, $\mathbf{D}_0$ is initialized to the all-zero vector, and for $t \geq 1$, $\mathbf{D}_t$ is obtained using $\mathbf{D}_{t-1}$ and $\mathbf{Y}_t$ (refer [26], [30] for the details). It can be seen that given $\mathbf{D}_0, \mathbf{D}_1, \dots, \mathbf{D}_{t-1}$, the random vector $\mathbf{D}_t$ depends only on $\mathbf{D}_{t-1}$ and hence the sequence $\mathbf{D}_0, \mathbf{D}_1, \dots, \mathbf{D}_N$ form a Markov chain (MC) of order one [31]. Further for the BSC, it is known that the number of values $\mathbf{D}_t$ can take are finite in number and thus the $\mathbf{D}_1^N$ follows a discrete-state, discrete-time, time-invariant MC [31].

Note that depending on whether $\mathbf{Y}_1^N$ follows the hypothesis H_1 or H_2 , the state transition matrices of MC corresponding to $\mathbf{D}_1^N$ will be different. Let P_j be the state transition matrix under hypotheses H_j , for $j = 1, 2$. For our code detection problem, we now use the statistic $\mathbf{D}_1^N$. The hypothesis testing of Eq. (1) is now reduced to the problem of hypothesis testing between the two MCs P_1 and P_2 given by

$$\begin{aligned} H_1 : \mathbf{D}_1^N &\sim \text{Markov Chain with } P_1 \\ H_2 : \mathbf{D}_1^N &\sim \text{Markov Chain with } P_2. \end{aligned} \quad (2)$$

For the given received data $\mathbf{Y}_1^N$, our proposed method for code detection is summarized below:

- Decode $\mathbf{Y}_1^N$ using code $\mathcal{C}_1$ via Viterbi algorithm to get the sequence of relative Viterbi metric $\mathbf{D}_1^N$.
- Find the state transition matrices P_1 and P_2 using the method that will be described in Section III-B.
- Using P_j , find the likelihood of $\mathbf{D}_1^N$ under hypotheses H_j , denoted by $\mathbb{P}[\mathbf{D}_1^N|H_j]$, for $j = 1, 2$.
- Hypothesis testing is performed as follows:

$$\frac{\mathbb{P}(\mathbf{D}_1^N|H_1)}{\mathbb{P}(\mathbf{D}_1^N|H_2)} \underset{H_2}{\overset{H_1}{\geq}} \tau, \quad (3)$$

where τ is the threshold. For our problem, since H_1 and H_2 are equally likely, $\tau = 1$ is the optimal threshold as per the Bayesian setup [28].

B. Finding the state transition matrices P_1 and P_2

We first discuss how to find P_1 , which involves finding the transition probabilities $\mathbb{P}[\mathbf{D}_t|\mathbf{D}_{t-1}]$ for all possible values $\mathbf{D}_t$ and $\mathbf{D}_{t-1}$ can take. Recall that under H_1 , the data $\mathbf{Y}_1^N$ is generated according to code $\mathcal{C}_1$ and we are decoding it according to code $\mathcal{C}_1$. To aid this discussion, we also explain the respective steps via Example 1 (see Fig. 2). A discussion towards finding P_1 can also be found in [31]. In order to find $\mathbb{P}[\mathbf{D}_t|\mathbf{D}_{t-1}]$, the key idea is to use the calculations done towards finding the relative Viterbi metric for the one step of trellis. We now focus on the section of the trellis from time $t-1$ to t . Suppose $\mathbf{D}_{t-1} = [D_{t-1,1} \ D_{t-1,2} \ \dots \ D_{t-1,2^m}]$, where note that each $D_{t-1,i}$ corresponds to a vertex in the trellis at time $t-1$. Similarly, we have $\mathbf{D}_t = [D_{t,1} \ D_{t,2} \ \dots \ D_{t,2^m}]$. The Viterbi algorithm finds $\mathbf{D}_t$ using $\mathbf{D}_{t-1}$ and $\mathbf{Y}_t$ as follows:

- For every vertex i in the trellis at time t , consider the set $\mathcal{S}_t(i)$ of incoming paths to this vertex. For every path $s \in \mathcal{S}_t(i)$, let $\mathbf{V}_t(s)$ denote the codeword corresponding to the path s . Similarly, let $D_{t-1,(s)}$ denote the relative Viterbi metric corresponding to the incoming vertex at time $t-1$.
- The Viterbi algorithm computes $D'_{t,i}$, for $i = 1, 2, \dots, 2^m$ as follows

$$D'_{t,i} = \min_{s \in \mathcal{S}_t(i)} \left\{ D_{t-1,(s)} + d_H(\mathbf{Y}_t, \mathbf{V}_t(s)) \right\}, \quad (4)$$

where $d_H(\mathbf{Y}_t, \mathbf{V}_t(s))$ denotes the Hamming distance between vectors $\mathbf{Y}_t$ and $\mathbf{V}_t(s)$.

- The relative Viterbi metric $D_{t,i}$ at vertex i is computed by subtracting the minimum metric among all $D'_{t,i}$ as

$$D_{t,i} = D'_{t,i} - \min_{j \in \{1, \dots, 2^m\}} D'_{t,j}. \quad (5)$$

We see from Eq. (5) that any entry in the relative metric $\mathbf{D}_t$ can take values in the range $\{0, 1, \dots, n\}$ since $0 \leq d_H(\mathbf{Y}_t, \mathbf{V}_t(s)) \leq n$. This implies that the support set of the random vector $\mathbf{D}_t$ can take finite set of values and these values can be obtained by enumerating over all possible values each entry $D_{t,i}$ can take. Let $\mathcal{D}$ be the support set of random vector $\mathbf{D}_t$. Note that $\mathbf{D}_t$'s are identically distributed but they are not independent. In order to compute $\mathbb{P}[\mathbf{D}_t|\mathbf{D}_{t-1}]$,

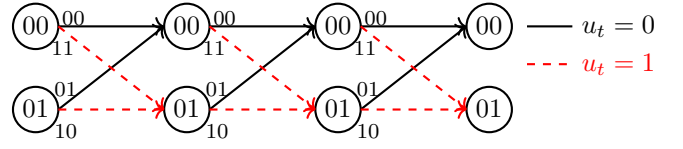


Fig. 2: Trellis for rate 1/2 code with $m = 1$ and generators $[1, 3]$ for $N = 3$ time steps

owing to the linearity property of convolutional codes, we can consider the situation when the all-zero codeword is transmitted [31]. For each value $\mathbf{d}_{t-1}$ that $\mathbf{D}_{t-1}$ can take, each distinct value of $\mathbf{y}_t \in \mathbb{F}_2^n$ will lead to a particular value $\mathbf{d}_t$ of $\mathbf{D}_t$, which can be computed using Eq. (4) and Eq. (5). Let $\mathcal{Y}(\mathbf{d}_{t-1}, \mathbf{d}_t)$ be the set of $\mathbf{y}_t \in \mathbb{F}_2^n$ that takes $\mathbf{D}_{t-1} = \mathbf{d}_{t-1}$ to $\mathbf{D}_t = \mathbf{d}_t$. The transition probability $\mathbb{P}[\mathbf{D}_t|\mathbf{D}_{t-1}]$ is then computed as

$$\begin{aligned} \mathbb{P}[\mathbf{D}_t = \mathbf{d}_t | \mathbf{D}_{t-1} = \mathbf{d}_{t-1}] &= \sum_{\mathbf{y}_t \in \mathcal{Y}(\mathbf{d}_{t-1}, \mathbf{d}_t)} \mathbb{P}[\mathbf{Y}_t = \mathbf{y}_t] \\ &\stackrel{(a)}{=} \sum_{\mathbf{y}_t \in \mathcal{Y}(\mathbf{d}_{t-1}, \mathbf{d}_t)} p^{w(\mathbf{y}_t)} (1-p)^{n-w(\mathbf{y}_t)}, \end{aligned} \quad (6)$$

where (a) in Eq. (6) is obtained since the all-zero codeword is transmitted and $w(\mathbf{y}_t)$ denotes the Hamming weight of $\mathbf{y}_t$.

In the following example, we illustrate the calculations towards finding P_1 via Eq. (6) for a simple example of a convolutional code with memory $m = 1$.

Example 1 (Rate-1/2 code with memory $m = 1$). Consider a convolutional code with generators $(1, 3)$ in octal notation. The trellis corresponding to this code for $N = 3$ time steps is shown in Fig. 2. The set of all possible relative metric $\mathbf{D}_t$ can take will be $\mathcal{D} = \{[0 \ 0], [0 \ 1], [1 \ 0], [0 \ 2], [2 \ 0]\}$. When $\mathbf{D}_{t-1} = [0 \ 0]$, from Eq. (4), Eq. (5), and Fig. 2, we get $\mathbf{D}_t = [0 \ 1]$ when $\mathbf{y}_t = [0 \ 0]$ and $\mathbf{y}_t = [0 \ 1]$. Thus, the set $\mathcal{Y}([0 \ 0], [0 \ 1]) = \{[0 \ 0], [0 \ 1]\}$. From Eq. (6) we have $\mathbb{P}[\mathbf{D}_t = [0 \ 1] | \mathbf{D}_{t-1} = [0 \ 0]] = (1-p)^2 + (p(1-p)) = 1-p$. All transition probabilities can be computed all possible every possible $\mathbf{D}_{t-1} \in \mathcal{D}$ in this manner to get the complete state transition matrix P_1 given by

$$P_1 = \begin{bmatrix} 0 & 1-p & p & 0 & 0 \\ 2p(1-p) & 0 & 0 & (1-p)^2 & p^2 \\ p^2 + (1-p)^2 & 0 & 0 & p(1-p) & p(1-p) \\ 2p(1-p) & 0 & 0 & (1-p)^2 & p^2 \\ p^2 + (1-p)^2 & 0 & 0 & p(1-p) & p(1-p) \end{bmatrix},$$

where the ordering the rows and columns of P_1 is according to the states $\{[0 \ 0], [0 \ 1], [1 \ 0], [0 \ 2], [2 \ 0]\}$. $\square$

We now find the state transition matrix P_2 under hypothesis H_2 . Recall that under H_2 , the data $\mathbf{Y}_1^N$ is generated according to code $\mathcal{C}_2$ but we are decoding it according to code $\mathcal{C}_1$. In this case, one cannot consider the situation when the all zero-codeword is transmitted. Here, we need to condition over all possible codewords $\mathbf{W}_1^N$ in $\mathcal{C}_2$ to obtain the respective

values of $\mathbb{P}(\mathbf{Y}_1^N)$ in Eq. (6). However, these computations would require the weight distribution of $\mathbf{W}_1^N$ and may become computationally prohibitive, in general. In this case, we numerically observe that P_2 can be approximated by substituting $p = 0.5$ in Eq. (6). This is motivated by the fact that mismatched decoding leads to transition statistics that resemble those of a maximally noisy (random) input. Finally, for a sequence $\mathbf{D}_1^N$, the likelihood is computed as $\mathbb{P}(\mathbf{D}_1^N | H_j) = \mathbb{P}(\mathbf{D}_0) \prod_{t=1}^N P_j(\mathbf{D}_{t-1}, \mathbf{D}_t)$.

C. Error exponents for hypothesis testing of Eq. (2)

To study the analytical performance of our code detection method proposed in Section III-A, we analyze the asymptotic behavior of the hypothesis testing problem of Eq. (2). The asymptotic performance of a hypothesis testing can be characterized by its *error exponent*, which measures the rate at which the probability of decision error goes to zero as the number of samples tend to infinity [27].

Let I_{err} denote the error exponent associated with Eq. (2). Note that this is a hypothesis testing problem between two discrete-state, discrete-time, time-invariant MCs with transition matrices P_1 and P_2 . For analytical derivation, consider the matrices P_1 and P_2 , and for $u \in [0, 1]$, define the matrix $M(u)_{i,k} = P_{1,i,k}^u P_{2,i,k}^{1-u}$. Let $\lambda(u)$ denote the spectral radius of $M(u)$. Then the asymptotic error exponent I_{err} is characterized by the optimization problem [28, Ch. 12]

$$I_{\text{err}} = - \min_{u \in [0,1]} \ln(\lambda(u)). \quad (7)$$

It is known that $\lambda(u)$ is a convex function of u [28, Sec. 12.2.3], and this convexity property enables efficient numerical computation of Eq. (7). For our code detection method, we find I_{err} analytically using Eq. (7) and also estimate it numerically (see Table II for an illustration).

IV. SIMULATIONS

In this section, we present simulation results comparing the proposed MC-based method with the syndrome-former baseline proposed by Moosavi *et al.* [2]. Python-based code implementations for both methods are available ². We evaluate performance under varying channel conditions and observation lengths. Specifically, Section IV-A summarizes the main findings and performance trends, Section IV-B analyzes the asymptotic behavior of both methods, and Section IV-C discusses key distinctions, including threshold sensitivity, analytical tractability, and computational considerations.

Originally proposed for AWGN channel, the syndrome-former method of [2] utilizes a syndrome-former equation satisfied by code-1 but not code-2. The probability that the received data satisfies this equation serves as the test statistic. For this work, we appropriately adapted these probability computations for BSC. Similar syndrome-former-based ideas are widely used in the literature for detecting other channel codes, such as LDPC codes [16], [24].

²<https://anonymous.4open.science/r/Detecting-Convolutional-Codes-Via-Markovian-Statistics-716B/>

For simulations, we focus on the following examples:

- Ex. 1: $m = 2$, rate-1/2, code (4, 6) vs (6, 7)
- Ex. 2: $m = 2$, rate-1/3, code (3, 2, 4) vs (3, 5, 4)
- Ex. 3: $m = 3$, rate-2/3, $\{(17, 13), (15, 11), (16, 12)\}$ vs $\{(12, 17), (15, 12), (14, 10)\}$
- Ex. 4: $m = 4$, rate-1/2, code (30, 26) vs (21, 20)

Note that we have chosen convolutional codes of various memories and rates.

A. Simulations for the probability of incorrect decision P_{err}

In the simulations, we plot the probability of incorrect decision P_{err} versus cross-over probabilities p and length N of the received data for our MC-based method and the baseline syndrome-former-based method. Fig. 3 summarizes the comparative performance for the four examples mentioned above. The top row shows P_{err} versus p , and the bottom row shows P_{err} versus N . In all plots, the solid line represents our method, while the dotted line represents the baseline. Different colors indicate different values for the BSC(p) values.

For Ex. 1, 2, and 3, the proposed method consistently achieves a lower P_{err} than the baseline, with P_{err} decaying toward zero at a significantly higher rate as N increases. Conversely, in Ex. 4, our performance is lower than the baseline; this occurs because Code-1 possesses 185 distinct relative-metric states, which makes the transition statistics under both hypotheses nearly identical and complicates the MC-based detection.

A detailed comparison for Ex. 2 is provided in Table I. At short observation lengths ($N = 50$), the proposed MC-based method achieves a reduction in P_{err} by a factor of 3 to 8 compared to the syndrome-former-based method across all tested noise levels. As N increases, this performance gap widens significantly. Notably, for $p = 0.2$ and $N = 500$, our proposed method achieves near-perfect detection ($P_{\text{err}} \approx 0$), while the baseline remains error-prone ($P_{\text{err}} \approx 0.41$). As expected, all curves in Fig. 3(a)-(d) converge to $P_{\text{err}} = 0.5$ at $p = 0.5$, consistent with the loss of discriminability in a maximally noisy channel. Similar trends are observed for Ex. 1 and 3.

TABLE I: P_{err} comparison across p and N for Ex. 2

N	Method	0.1	0.15	0.2
50	Proposed (MC-based)	0.032	0.105	0.221
	Syndrome-former method	0.274	0.407	0.496
200	Proposed (MC-based)	0.000	0.005	0.065
	Syndrome-former method	0.163	0.396	0.487
500	Proposed (MC-based)	0.000	0.000	0.000
	Syndrome-former method	0.032	0.382	0.413

B. Asymptotic analysis via error exponents

In this subsection, we analyze the asymptotic performance of the two methods. Toward this end, we study the error exponent I_{err} which quantifies the rate at which the probability of decision error decays with the observation length N . For our method, we find the error exponent analytically using Eq. (7)

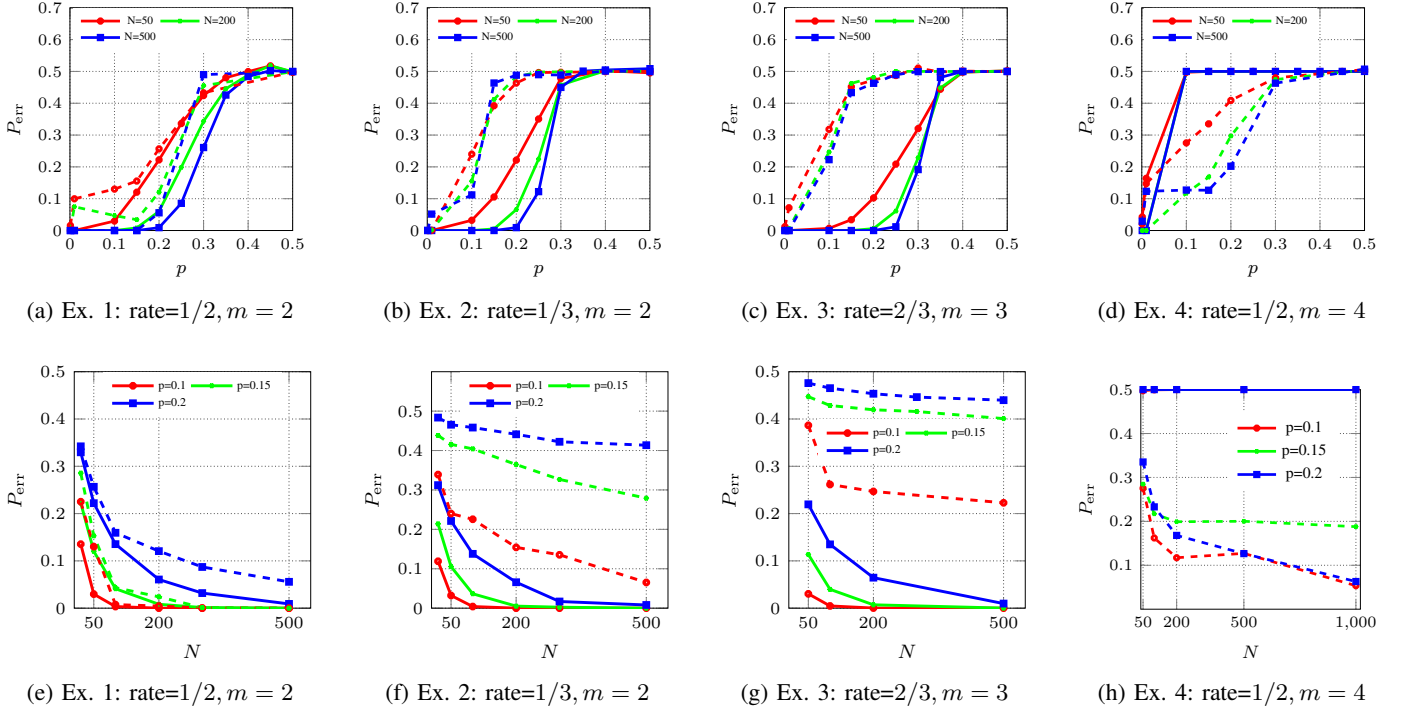


Fig. 3: (a)–(d): Probability of error P_{err} vs. BSC crossover probability p . (e)–(h): P_{err} vs. observation length N . Solid lines represent our method; dotted lines represent the baseline.

and empirically estimate it using the simulated P_{err} curves. Let $I_{\text{err}}(\text{emp})$ denote the error exponent estimated using the simulated curves. Empirically, $I_{\text{err}}(\text{emp})$ is estimated by fitting an exponential curve to the simulated values of $P_{\text{err}}(N)$, based on the model $P_{\text{err}}(N) \approx e^{-I_{\text{err}}(\text{emp})N}$. We also estimate similar error exponents for the baseline method, denoted by $I_{\text{err}}^{\text{Lit}}(\text{emp})$.

In Table II, we compare I_{err} , $I_{\text{err}}(\text{emp})$, and $I_{\text{err}}^{\text{Lit}}(\text{emp})$ for Ex. 2. The empirical estimates of the error exponent $I_{\text{err}}(\text{emp})$ obtained from simulation-based fits closely follow I_{err} obtained analytically. In addition, the proposed method shows higher empirical $I_{\text{err}}(\text{emp})$ values than the adapted baseline, indicating a faster asymptotic decay of the detection error probability with increasing N .

TABLE II: Comparison of error exponents for Ex. 2

p	I_{err} (via Eq. (7))	$I_{\text{err}}(\text{emp})$	$I_{\text{err}}^{\text{Lit}}(\text{emp})$
0.10	0.0209	0.0392	0.0197
0.15	0.0057	0.0177	0.0158
0.20	0.0012	0.0073	0.0029

C. Key distinctions with respect to the literature

We now compare the proposed MC-based method and the based syndrome-former method [2] on some key aspects.

1) *Sensitivity to threshold computation for hypothesis testing:* The proposed method utilizes a likelihood ratio test with threshold $\tau = 1$, which is the optimal choice for equally likely hypotheses. In contrast, the baseline method lacks an

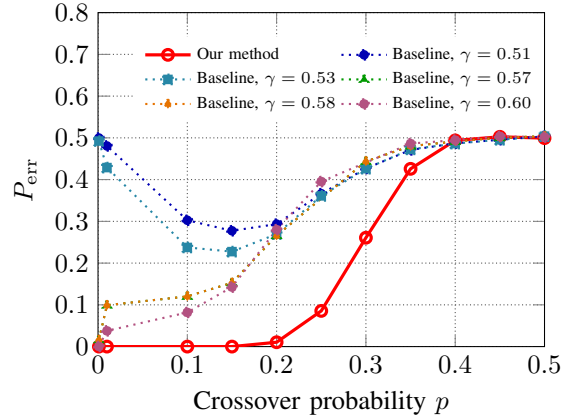


Fig. 4: P_{err} versus p for different threshold values γ in the baseline (dotted curves) for $R = 1/2$ code with $m = 2$. The proposed method (solid red) has threshold $\tau=1$.

analytical threshold derivation and relies on an empirically tuned boundary γ [2]. As illustrated in Fig. 4, the baseline performance is highly sensitive to the selection of γ , whereas our method consistently achieves optimal results with $\tau = 1$.

2) *Analytical characterization:* A key advantage of the proposed framework lies in its analytical tractability. For our method, the asymptotic error exponent can be computed by Eq. (7), enabling the analytical characterization of the detection performance. This form of analytical characterization is not available for the baseline, which must be assessed solely

through simulations.

3) *Computational considerations*: The primary limitation of the proposed method arises from the growth of the state space of Markov-chain with encoder memory m . Although the number of states remain tractable for moderate memory lengths, the complexity can increase rapidly for higher-memory encoders. However, we observe that the resulting state-transition matrices are typically sparse, with several transitions exhibiting negligible probabilities. In the future, we plan to devise techniques such as state aggregation or pruning to limit the growth of the state-space which may provide an avenue for future computational tractability.

V. CONCLUSION AND FUTURE WORK

In this paper, we introduced a novel Markov chain-based framework to detect convolutional codes when the noise is introduced through the binary symmetric channel. For the problem of detecting between two convolutional codes, we considered the Viterbi metric obtained by decoding the received data via either of the codes as a statistic for solving the underlying detection problem. For the asymptotic performance analysis of the proposed method, we studied the associated error exponents. For the general problem of blind identification of convolutional codes, we believe that we have introduced a new Markovian framework, which could be interesting on its own. While in the current paper, we focus on feedforward convolutional codes, we believe our algorithm can be generalized to codes with feedback too.

Future work includes: (i) extending the proposed Markovian formulation to other channels such as the additive white Gaussian noise channel and binary erasure channel, (ii) generalizing our blind code detection method beyond the pairwise identification, and (iii) extending analysis to the punctured convolutional codes.

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